FINITE TYPE OVER A LOCALLY NOETHERIAN BASE IMPLIES FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The two parts are handled in one proof environment; part (2) is a one-line consequence of part (1) and is stated as such by the author.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The source-and-target presentation of a locally of finite type morphism. Lemma lemma-local-source-target, used at proof lines 5802-5805. Produces the scheme morphism h and the surjective etale a.
- The scheme case: locally of finite type over a locally Noetherian base is locally of finite presentation. Morphisms, Lemma morphisms-lemma-noetherian-finite-type-finite-presentation, used at proof lines 5805-5808. Gives local finite presentation of h.
- Local finite presentation is read off from a source-and-target presentation. used at proof lines 5808-5809. Concludes part (1) for f.
- Finite type plus quasi-separated means quasi-compact and quasi-separated. used at proof lines 5811-5812. Gives part (2) from part (1).

Source reference. Original lemma 03MJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4751–4751. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01TX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2390–2390. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S.*

(1) *If Y is locally Noetherian and f locally of finite type then f is locally of finite presentation.*
(2) *If Y is locally Noetherian and f of finite type and quasi-separated then f is of finite presentation.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Assume $f : X \to Y$ locally of finite type and Y locally Noetherian. This means there exists a diagram as in Lemma 03MJ with h locally of finite type and surjective vertical arrow a. By Morphisms, Lemma 01TX h is locally of finite presentation. Hence $X \to Y$ is locally of finite presentation by definition. This proves (1). If f is of finite type and quasi-separated then it is also quasi-compact and quasi-separated and (2) follows immediately. □